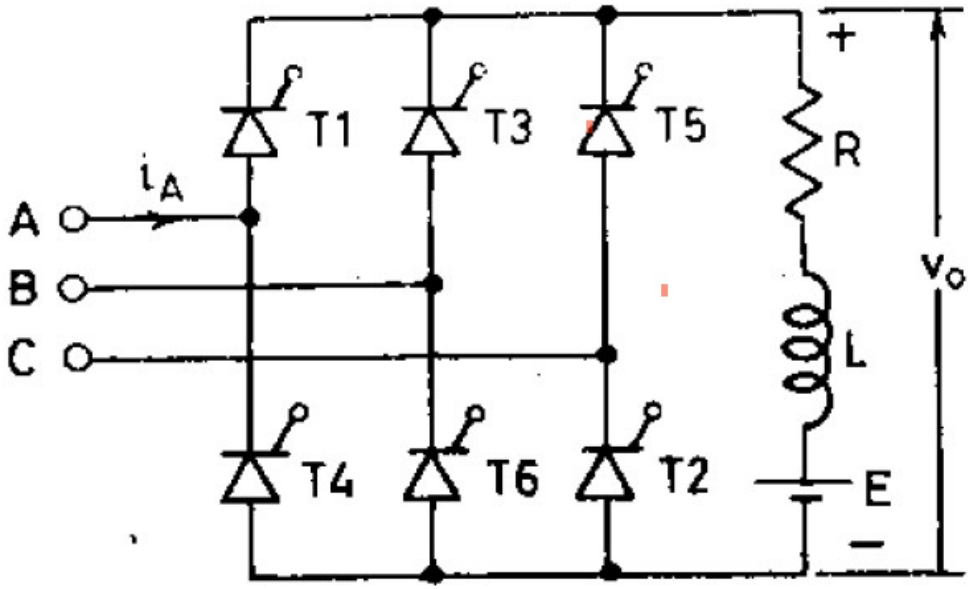
Phase controlled rectifiers

Three phase full converter

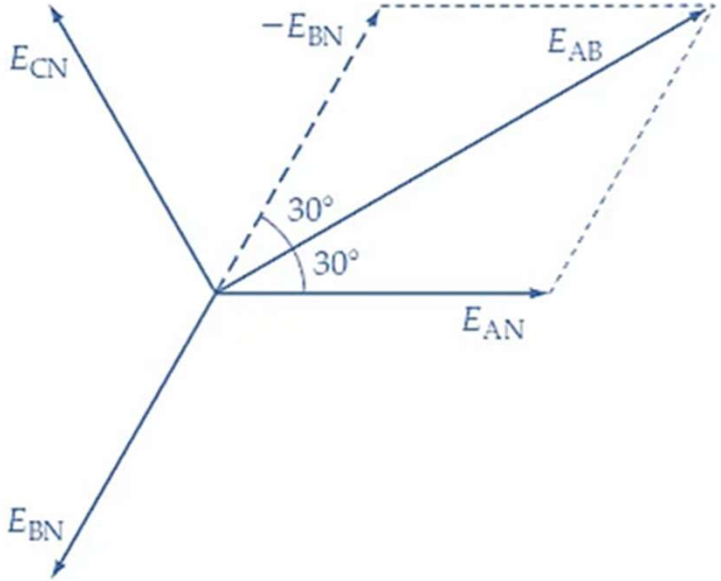


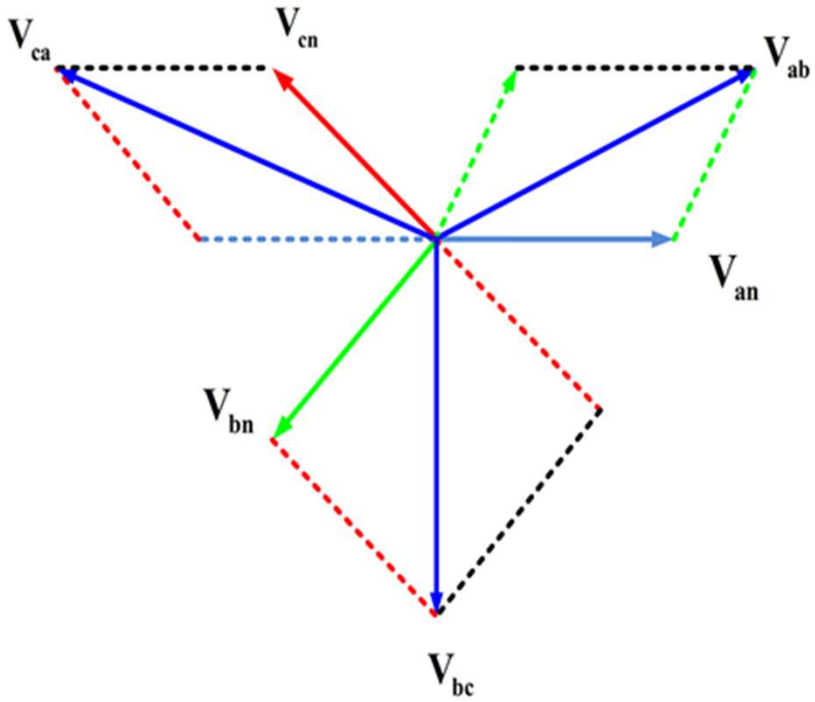
Converter or rectifier for

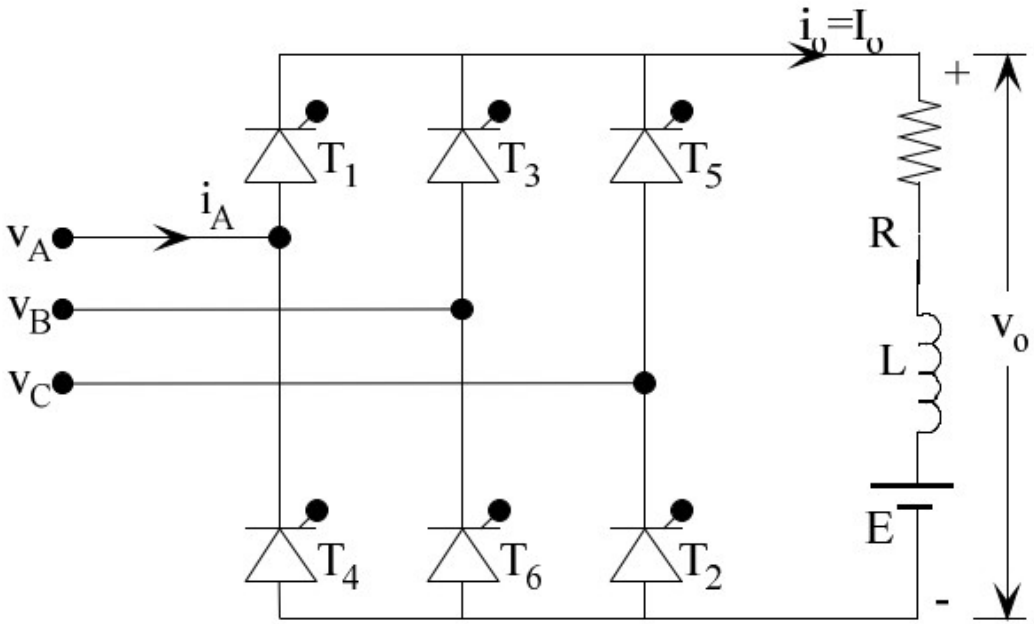
$0° < \alpha \leq 90°$

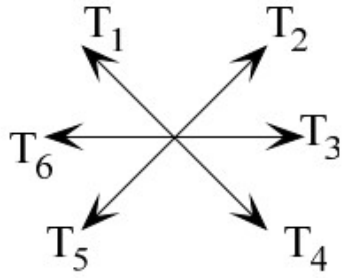
Inverter for

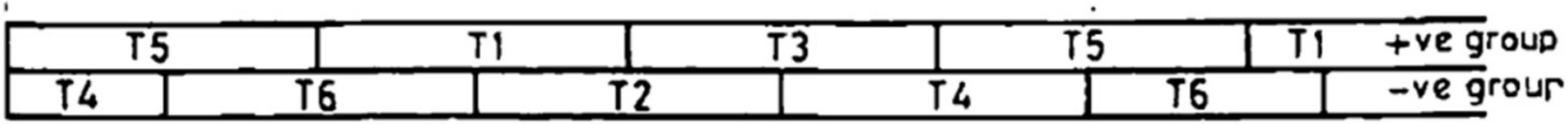
$90° < \alpha < 180°$

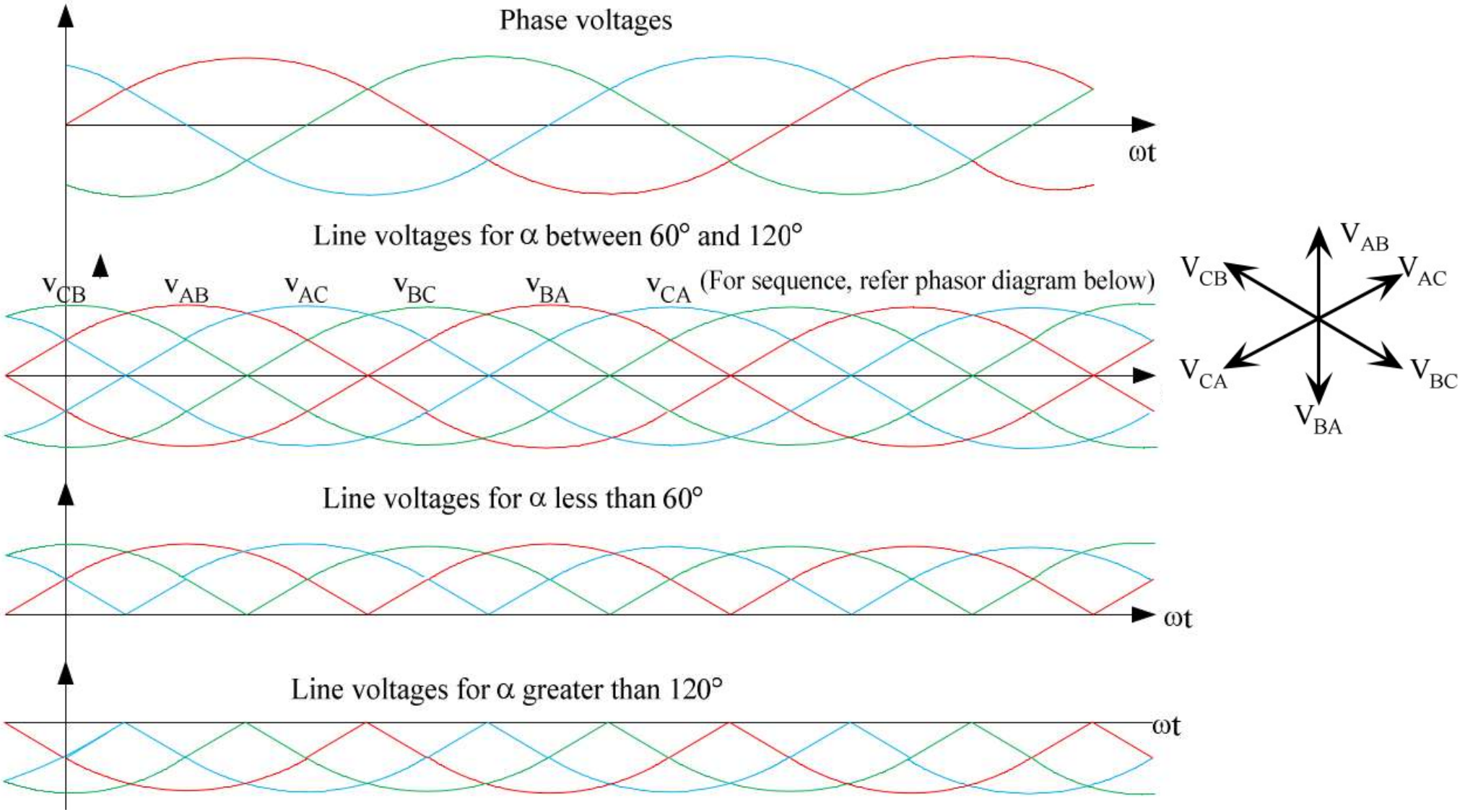

E_{CN}
$-E_{BN}$
E_{AB}
30°
30°
E_{AN}
E_{BN}


V_{ca}
V_{cn}
V_{ab}
V_{an}
V_{bn}
V_{bc}


$i_o = I_o$
+
T_1
T_3
T_5
i_A
v_A
v_B
v_C
R
L
E
v_o
-
T_4
T_6
T_2


T_1
T_2
T_6
T_3
T_5
T_4


T5
T1
T3
T5
T1
+ve group
T4
T6
T2
T4
T6
-ve group

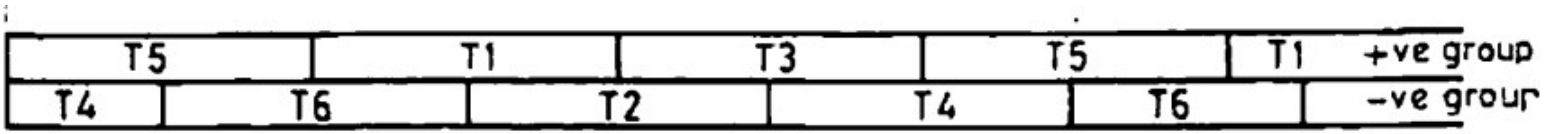

Phase voltages
ωt
Line voltages for α between 60° and 120°
v_{CB} v_{AB} v_{AC} v_{BC} v_{BA} v_{CA} (For sequence, refer phasor diagram below)
V_{AB}
V_{CB}
V_{AC}
V_{CA}
V_{BC}
V_{BA}
Line voltages for α less than 60°
ωt
Line voltages for α greater than 120°
ωt

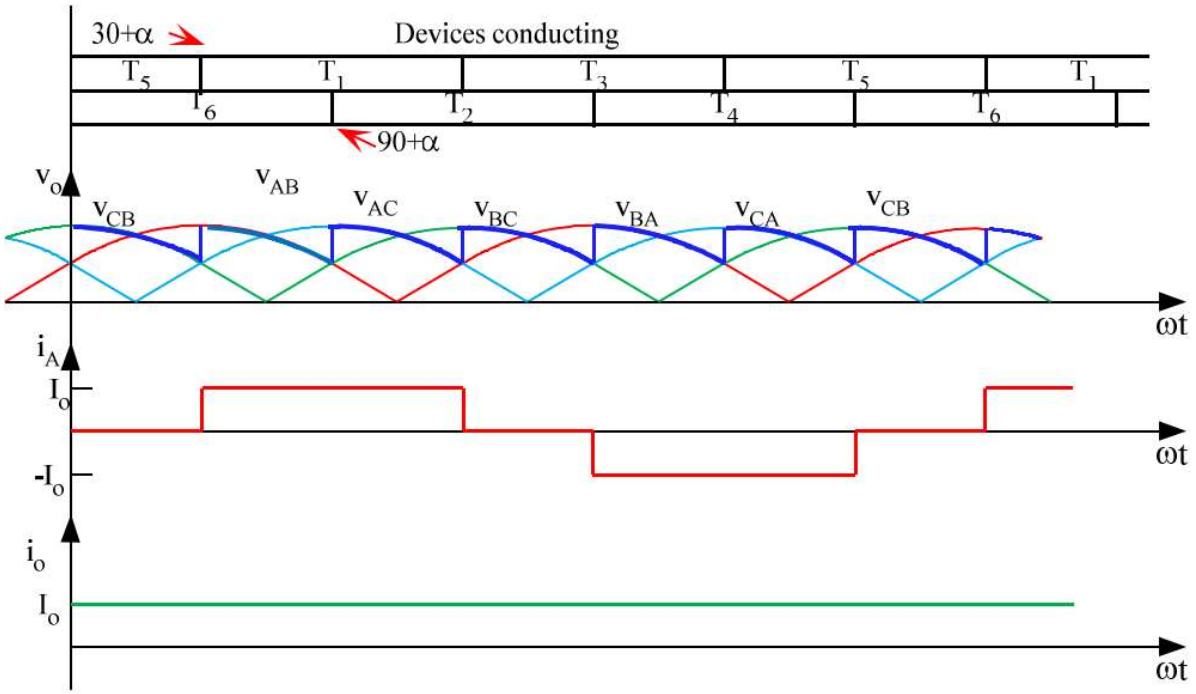
Device	Instant of turning ON	Instant of turning OFF
T_1	$\alpha+30°$	$\alpha+30°+120°$
T_2	$\alpha+90°$	$\alpha+90°+120°$
T_3	$\alpha+150°$	$\alpha+150°+120°$
T_4	$\alpha+210°$	$\alpha+210°+120°$
T_5	$\alpha+270°$	$\alpha+270°+120°$
T_6	$\alpha+330°$	$\alpha+330°+120°$

- In continuous conduction mode, each device will be conducting for 120°.
- T1 will be in forward blocking mode when its anode voltage is maximum positive which occurs after wt=30 .
- Hence, T1 can be triggered only after wt=30

- Similarly, T2, T3, T4, T5 and T6 can be triggered after wt=90, , 150, 210, 270, 330respectively.

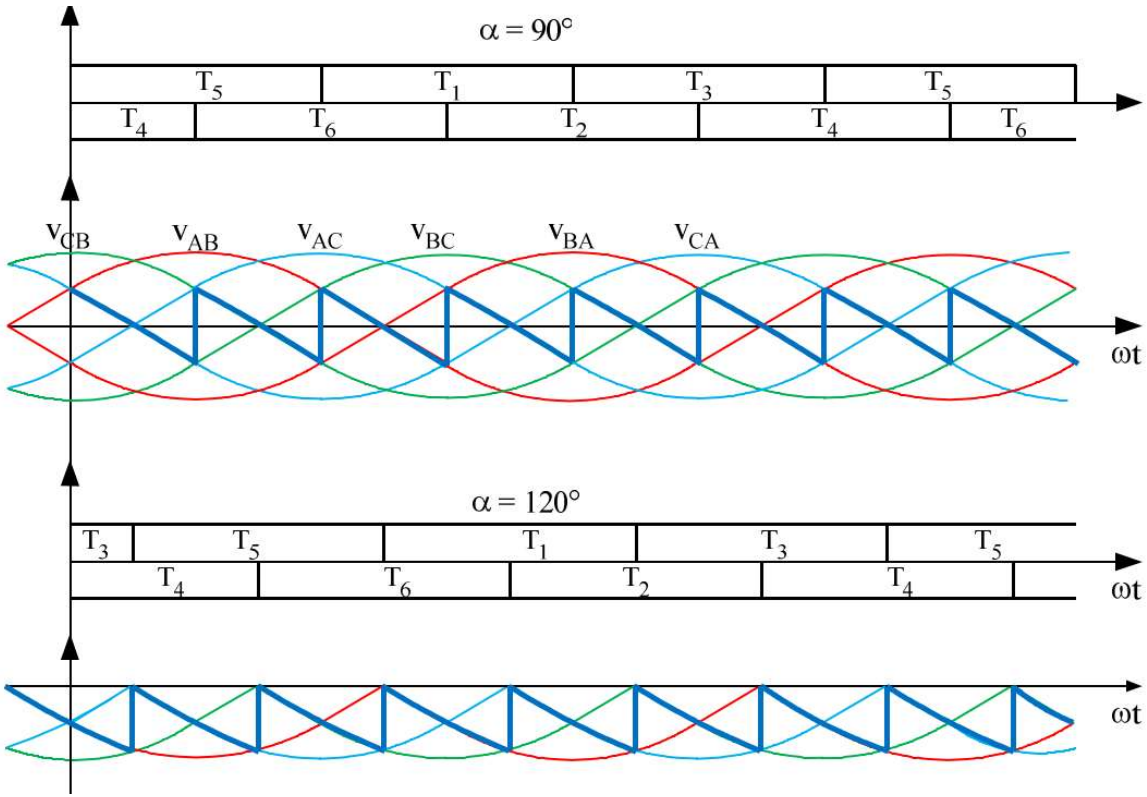
If firing angle is α, each device is fired in the following sequence.
T1 will conduct for 120° and it is turned off automatically
when T3 is fired.

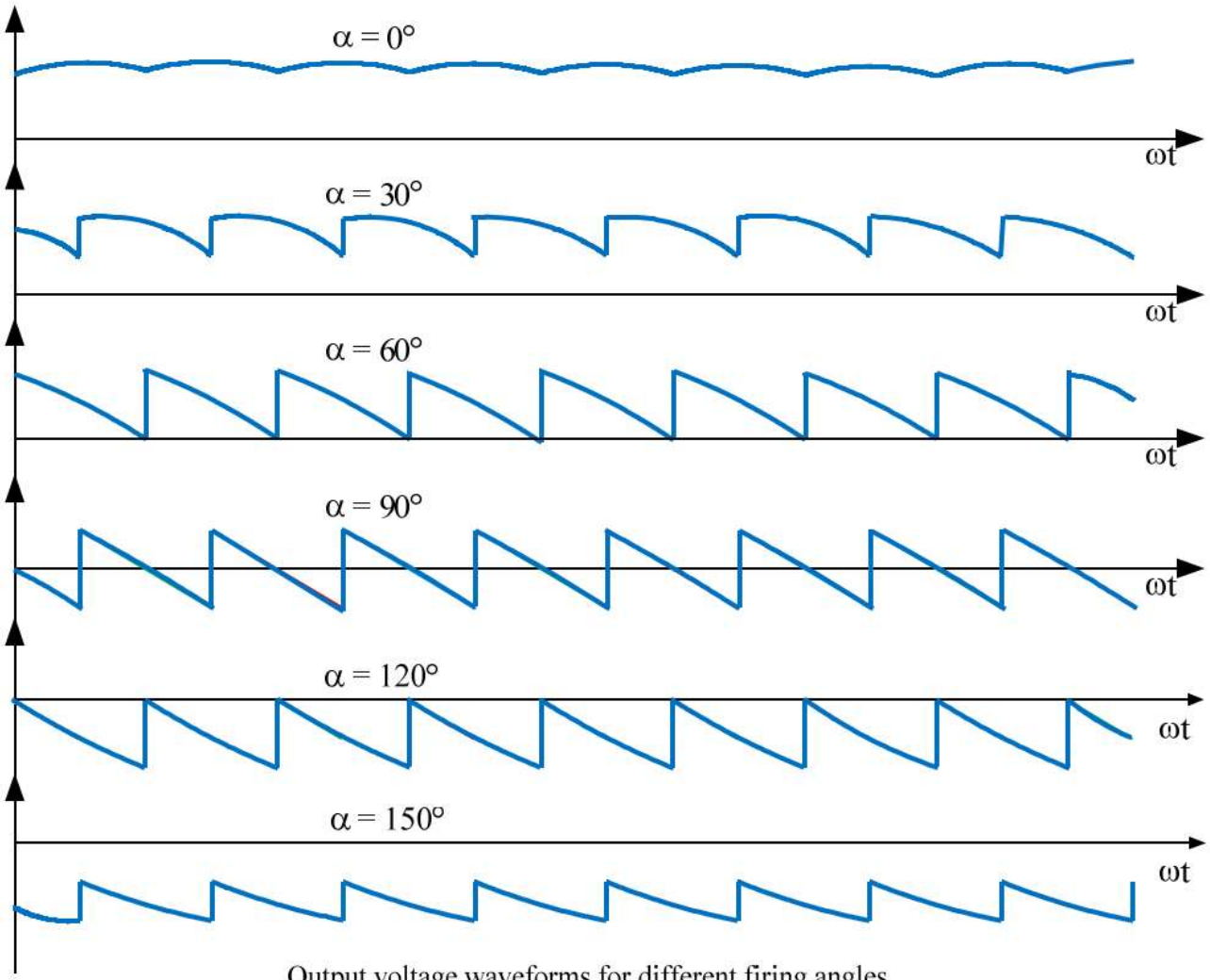




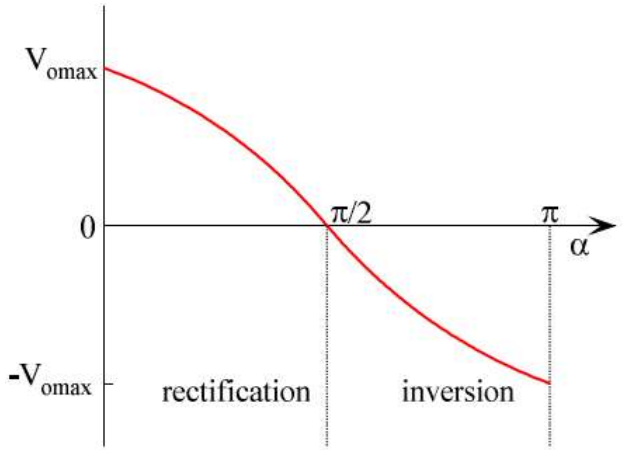
Device	Instant of turning ON	Instant of turning OFF
T_1	60°	180°
T_2	120°	240°
T_3	180°	300°
T_4	240°	360°
T_5	300°	420°
T_6	360°	480°

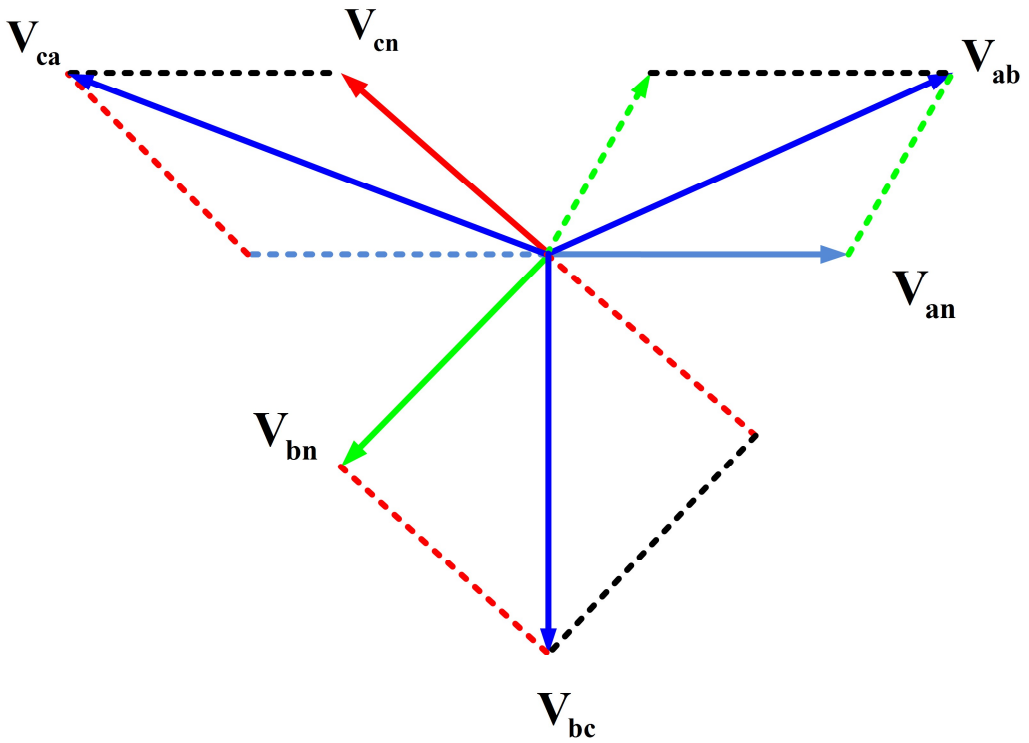
Duration	Devices ON	Load voltage
0-60°	T_5T_6	v_{CB}
60°-120°	T_1T_6	v_{AB}
120°-180°	T_1T_2	v_{AC}
180°-240°	T_3T_2	v_{BC}
240°-300°	T_3T_4	v_{BA}
300°-360°	T_5T_4	v_{CA}

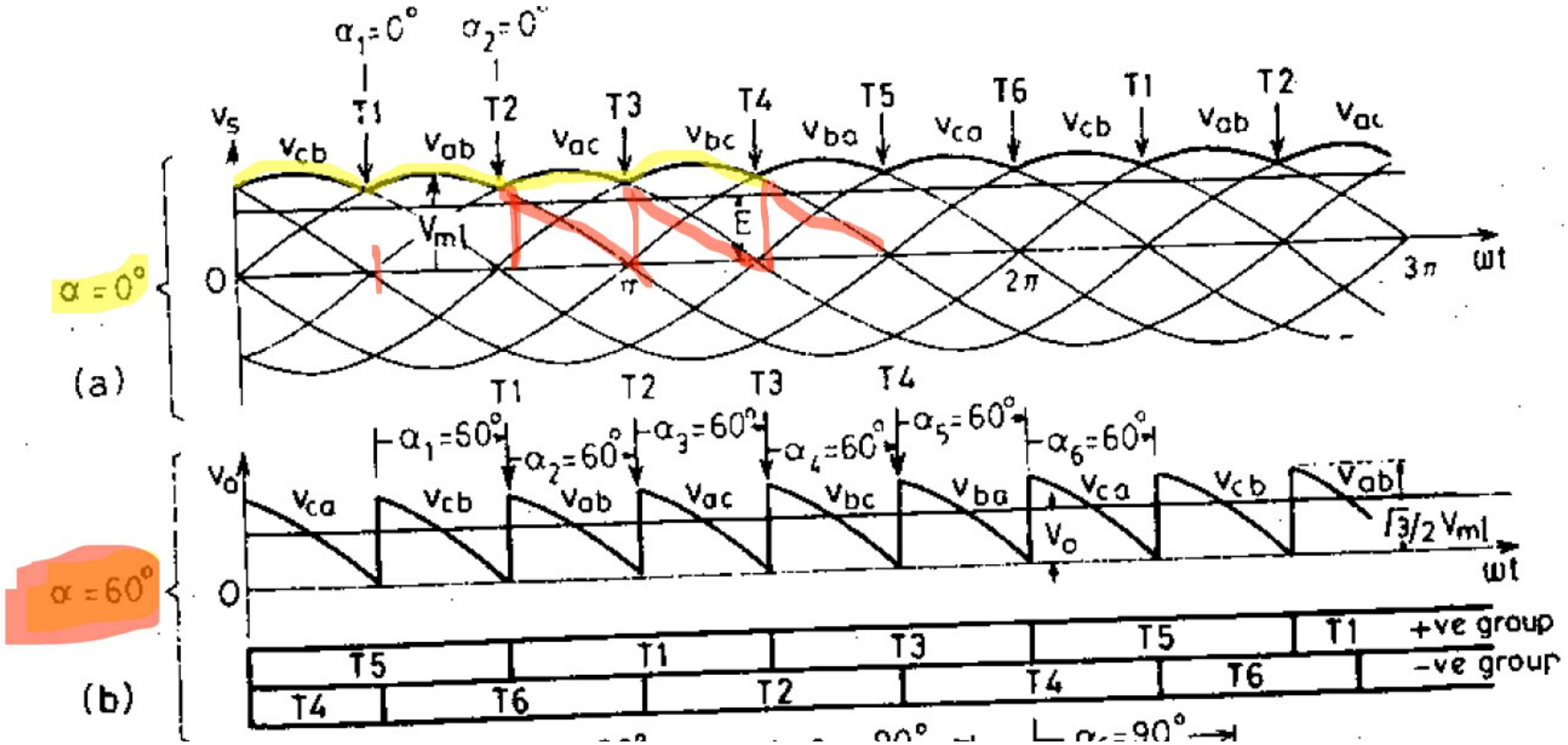

α = 90°
T5
T1
T3
T5
T4
T6
T2
T4
T6
vCB
vAB
vAC
vBC
vBA
vCA
ωt
α = 120°
T3
T5
T1
T3
T5
T4
T6
T2
T4
ωt
ωt

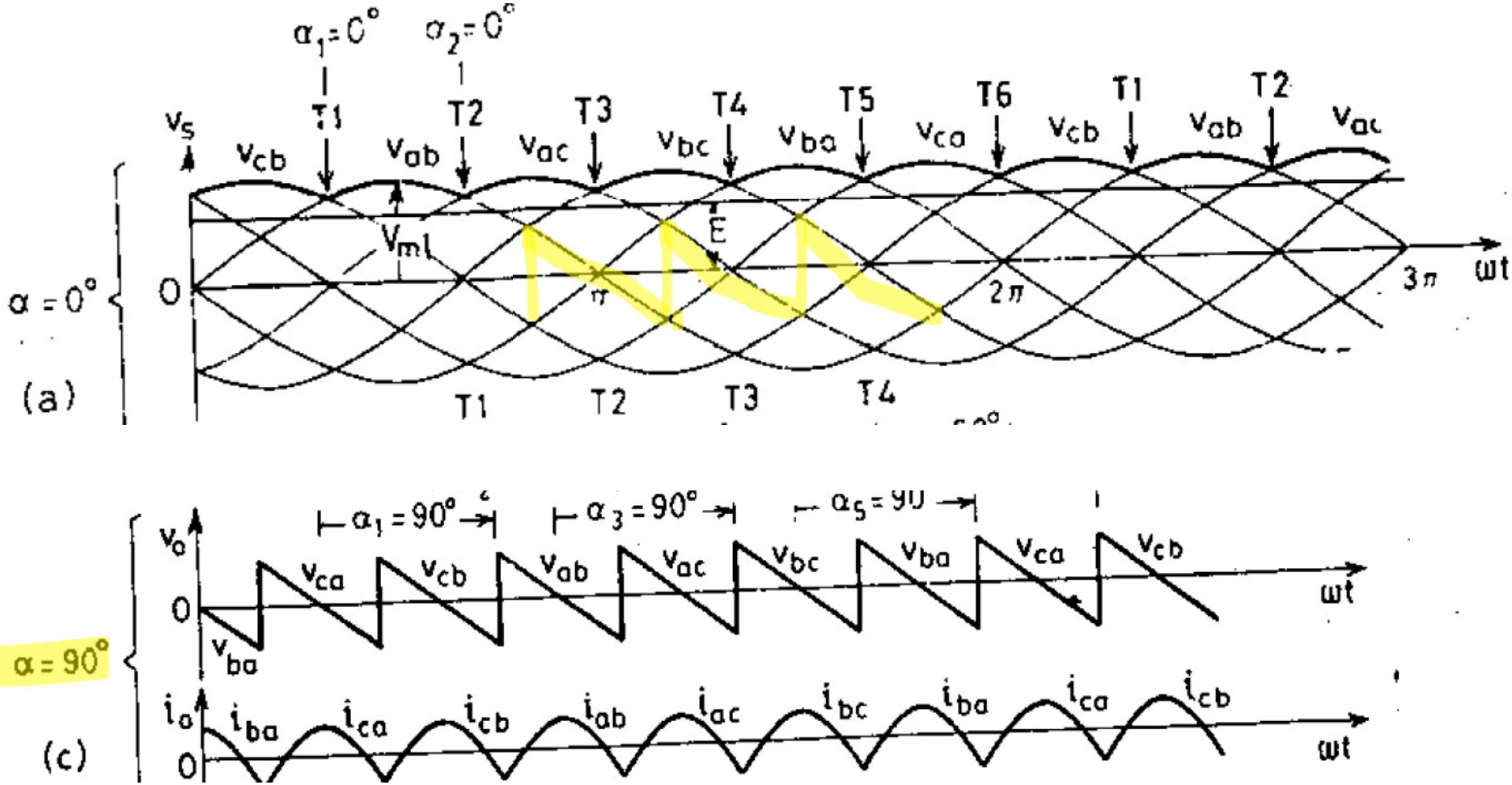


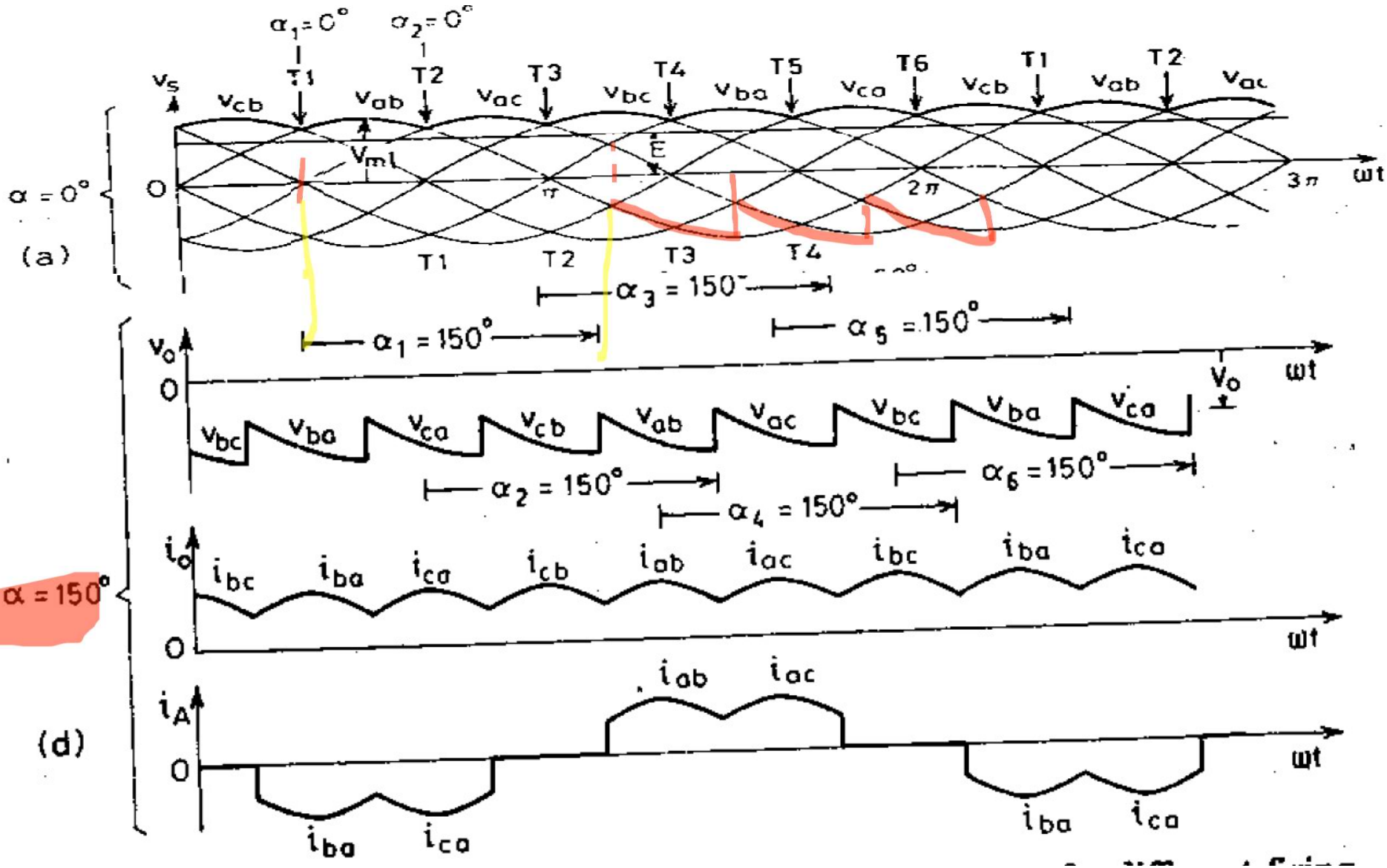
Output voltage waveforms for different firing angles

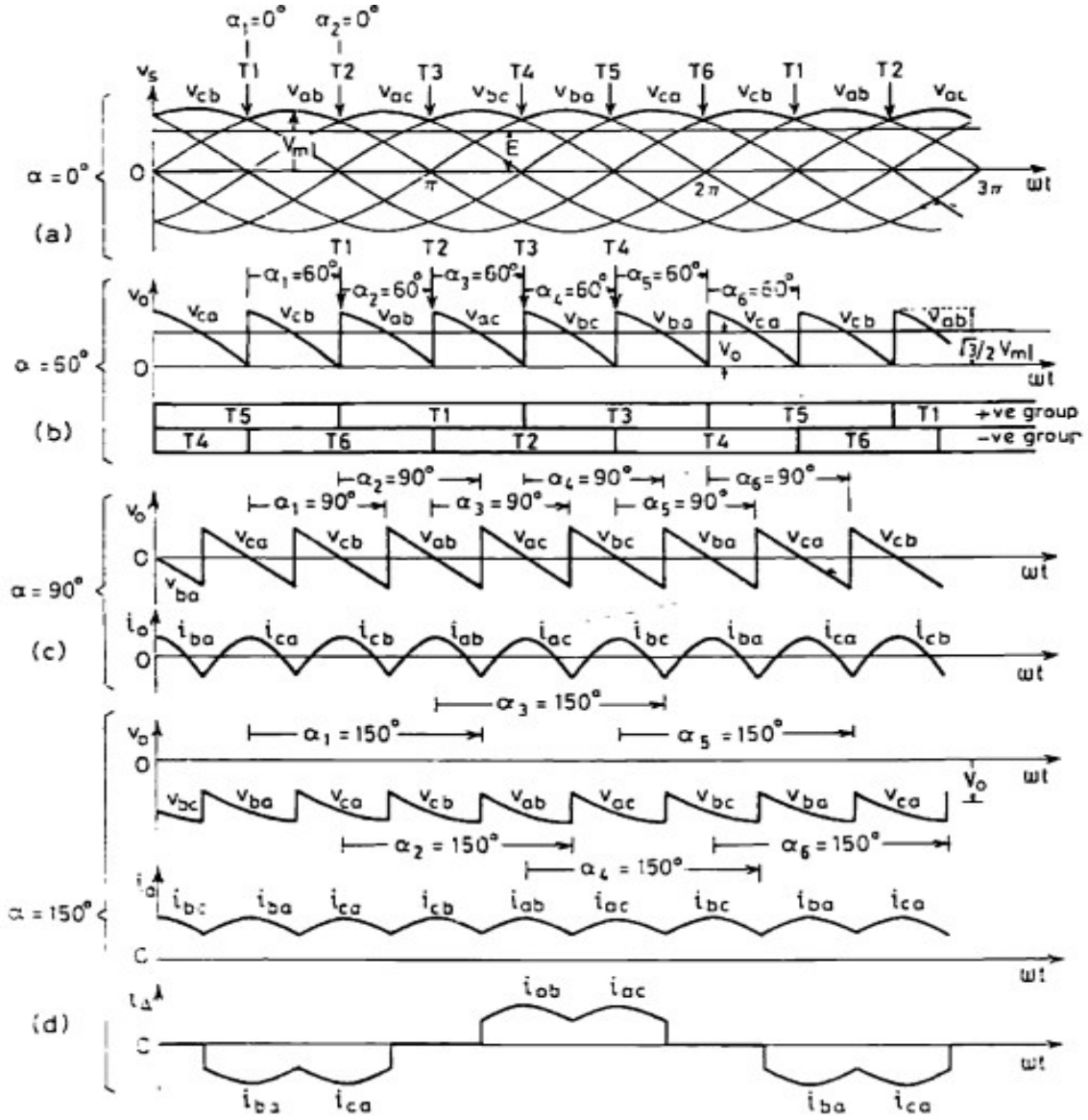


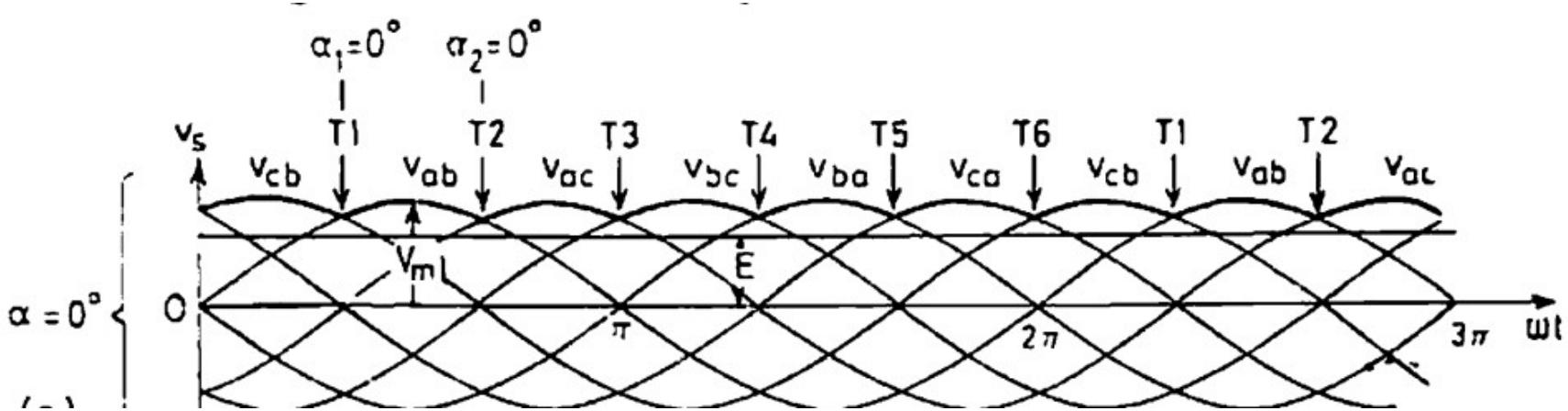

V_{ca}
V_{cn}
V_{ab}
V_{an}
V_{bn}
V_{bc}

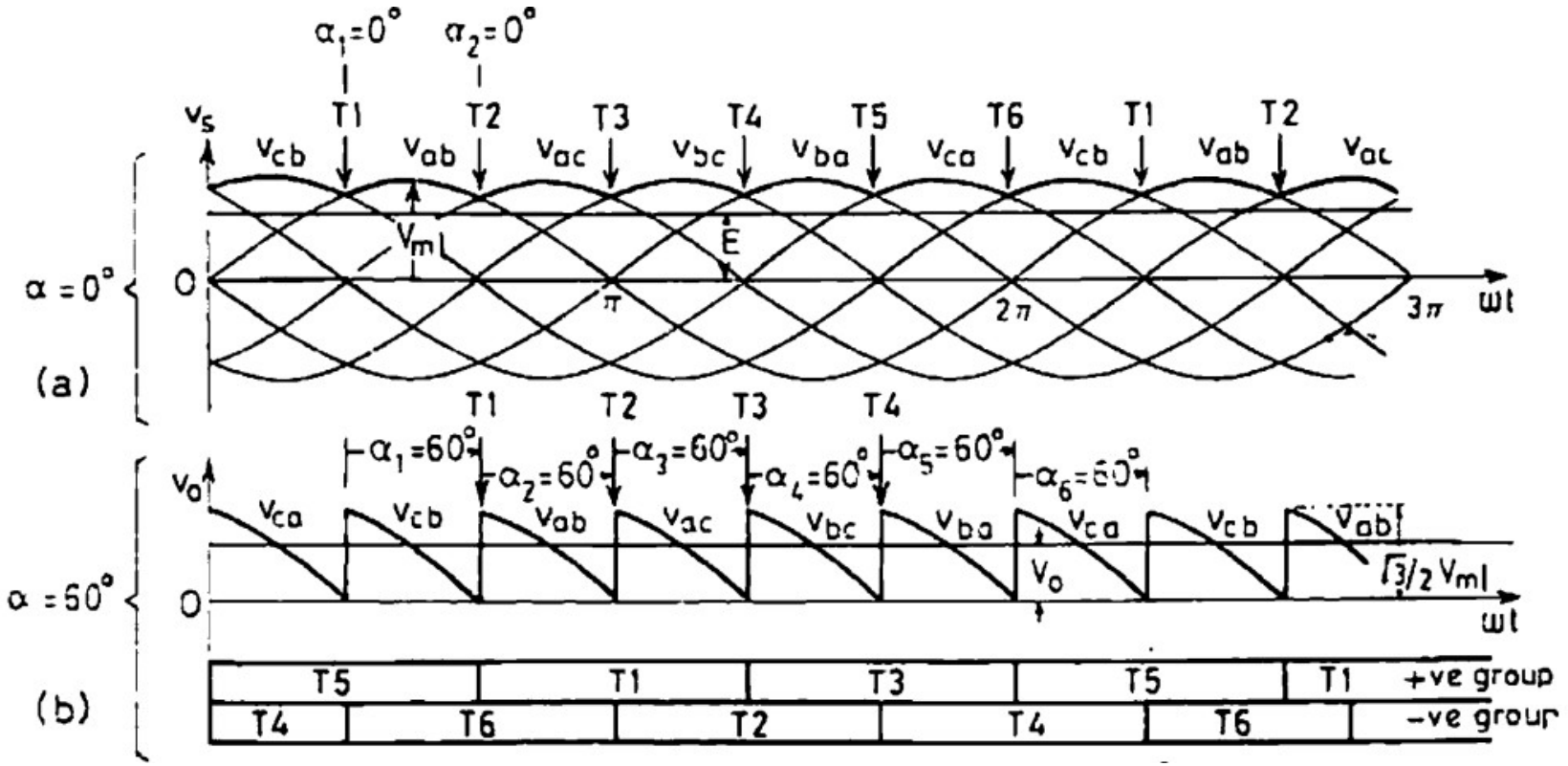

α = 0°
(a)
α = 60°
(b)
T1
T2
T3
T4
T5
T6
Vm
E
ωt
π
2π
3π
Vo
+ve group
-ve group
α1=60°
α2=60°
α3=60°
α4=60°
α5=60°
α6=60°
√3/2 Vm


α1 = 0°
α2 = 0°
T1
T2
T3
T4
T5
T6
T1
T2
vs
vcb
vab
vac
vbc
vba
vca
vcb
vab
vac
Vml
E
0
π
2π
3π
ωt
α = 0°
(a)
T1
T2
T3
T4
α1 = 90°
α3 = 90°
α5 = 90
vo
vca
vcb
vab
vac
vbc
vba
vca
vcb
vba
ωt
α = 90°
io
iba
ica
icb
iab
iac
ibc
iba
ica
icb
ωt
(c)

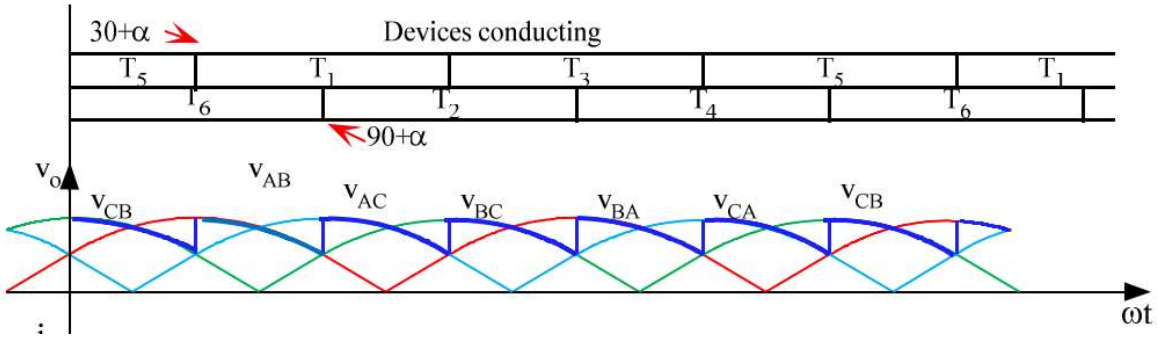

α = 0°
(a)
$\alpha_1 = 0°$
$\alpha_2 = 0°$
T1
T2
T3
T4
T5
T6
T1
T2
v_s
v_{cb}
v_{ab}
v_{ac}
v_{bc}
v_{ba}
v_{ca}
v_{cb}
v_{ab}
v_{ac}
V_{ml}
E
0
π
2π
3π
ωt
T1
T2
T3
T4
$\alpha_3 = 150°$
$\alpha_1 = 150°$
$\alpha_5 = 150°$
v_o
0
V_o
ωt
v_{bc}
v_{ba}
v_{ca}
v_{cb}
v_{ab}
v_{ac}
v_{bc}
v_{ba}
v_{ca}
$\alpha_2 = 150°$
$\alpha_6 = 150°$
$\alpha_4 = 150°$
α = 150°
i_o
i_{bc}
i_{ba}
i_{ca}
i_{cb}
i_{ab}
i_{ac}
i_{bc}
i_{ba}
i_{ca}
0
ωt
(d)
i_A
i_{ab}
i_{ac}
0
ωt
i_{ba}
i_{ca}
i_{ba}
i_{ca}

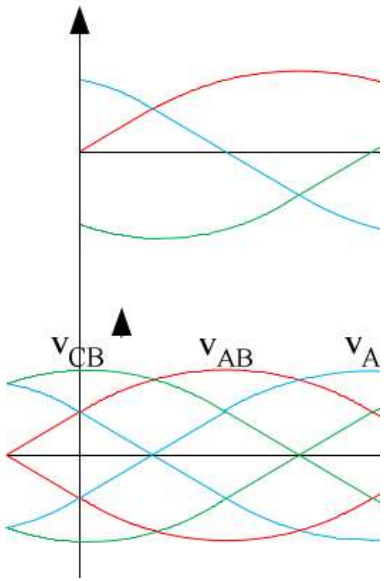

α = 0°
(a)
α = 60°
(b)
α = 90°
(c)
α = 150°
(d)
T1
T2
T3
T4
T5
T6
+ve group
-ve group
ωt


α1=0°
α2=0°
vs
T1
T2
T3
T4
T5
T6
T1
T2
vcb
vab
vac
vbc
vba
vca
vcb
vab
vac
Vml
E
α = 0°
O
π
2π
3π
ωt


α1=0°
α2=0°
vs
T1
T2
T3
T4
T5
T6
T1
T2
vcb
vab
vac
vbc
vba
vca
vcb
vab
vac
Vml
E
α = 0°
O
π
2π
3π
ωt
(a)
T1
T2
T3
T4
α1=60°
α2=60°
α3=60°
α4=60°
α5=60°
α6=60°
vo
vca
vcb
vab
vac
vbc
vba
vca
vcb
vab
Vo
√3/2 Vml
α = 60°
O
ωt
(b)
T5
T1
T3
T5
T1
+ve group
T4
T6
T2
T4
T6
-ve group

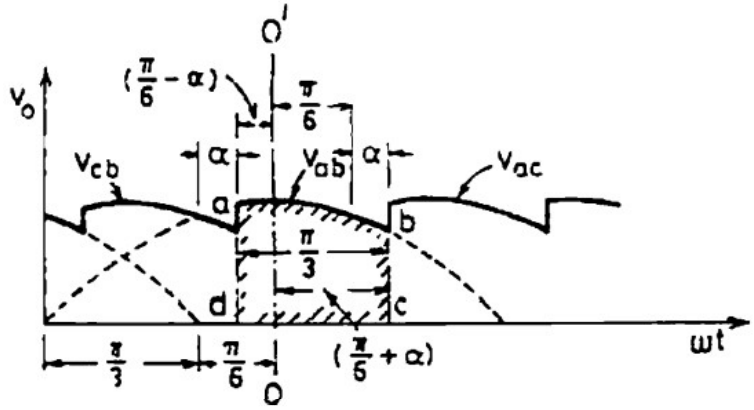
AVERAGE OUTPUT VOLTAGE of 6 pulse converter - Continuous Conduction





Using line voltage reference

If sine function is used for the source voltage, then $v_{ab} = V_{ml} \sin \omega t$ because $v_{ab} = 0$ at $\omega t = 0$.

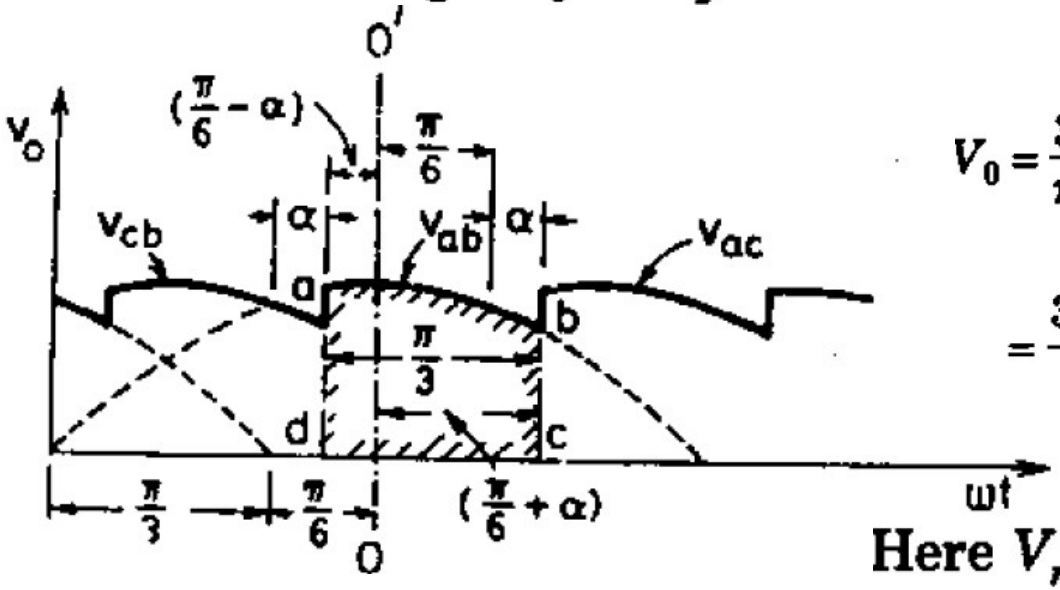


$$V_0 = \frac{3}{\pi} \int_{\frac{\pi}{3}+\alpha}^{\frac{2\pi}{3}+\alpha} V_{ml} \sin \omega t \cdot d(\omega t)$$

$$= -\frac{3V_{ml}}{\pi} \left[\cos\left(\frac{2\pi}{3} + \alpha\right) - \cos\left(\frac{\pi}{3} + \alpha\right) \right] = \frac{3V_{ml}}{\pi} \cos \alpha$$

Using line voltage reference – alternate method

firing angle delay $\alpha < 30°$. Note that periodicity of output voltage is $\pi/3$ radians. Average value of output voltage is obtained by finding the dashed area *abcd* over a periodic cycle, and then dividing it by the periodic time.



$$V_0 = \frac{3}{\pi}\int_{-\left(\frac{\pi}{6}-\alpha\right)}^{\left(\frac{\pi}{6}+\alpha\right)} V_{ml}\cos\omega t \cdot d(\omega t)$$

$$= \frac{3V_{ml}}{\pi}\left[\sin\left(\alpha+\frac{\pi}{6}\right) - \sin\left(\alpha+\frac{\pi}{6}\right)\right] = \frac{3V_{ml}}{\pi}\cos\alpha$$

Here V_{ml} is the maximum value of line voltage.

Source current

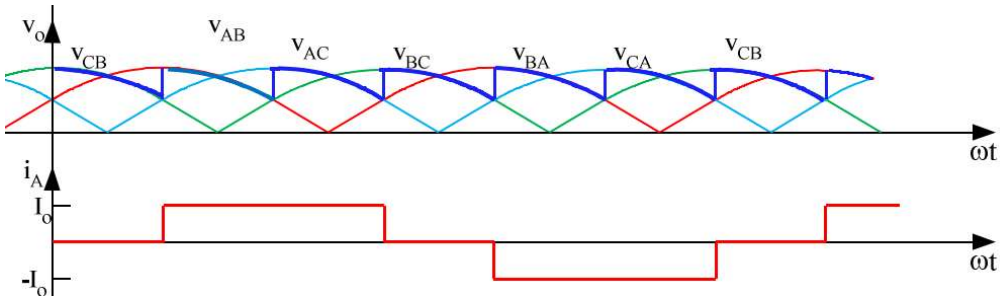
source current for phase A, *i.e.* i_A (or for any other phase)

flows for 120° for every 180°.

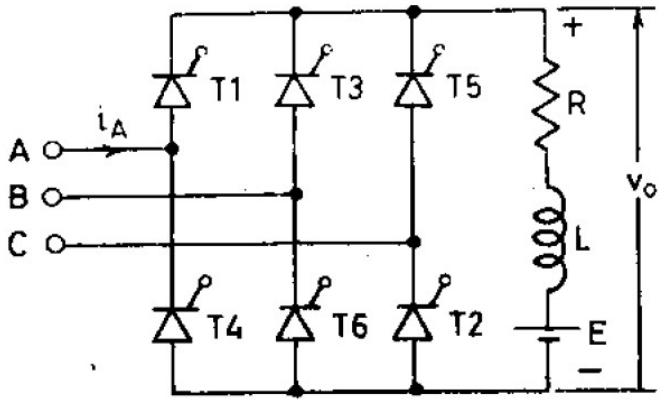
in case output current is assumed constant at I_0, the

RMS Value

$$I_s = \sqrt{I_0^2 \frac{2\pi}{3} \times \frac{1}{\pi}} = I_0 \sqrt{\frac{2}{3}}$$



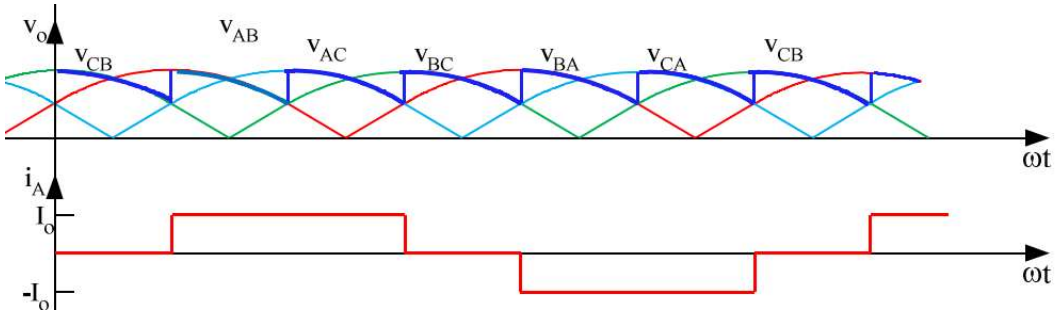
Thyristor current



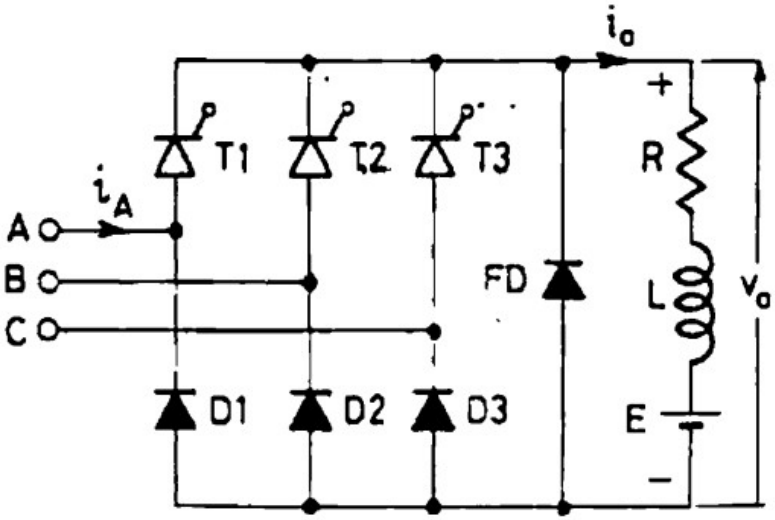
Each SCR conducts for 120° for every 360°.

Therefore, the rms value of thyristor current

$$I_{Th} = \sqrt{I_0^2 \frac{2\pi}{3} \times \frac{1}{2\pi}} = I_0 \sqrt{\frac{1}{3}}$$

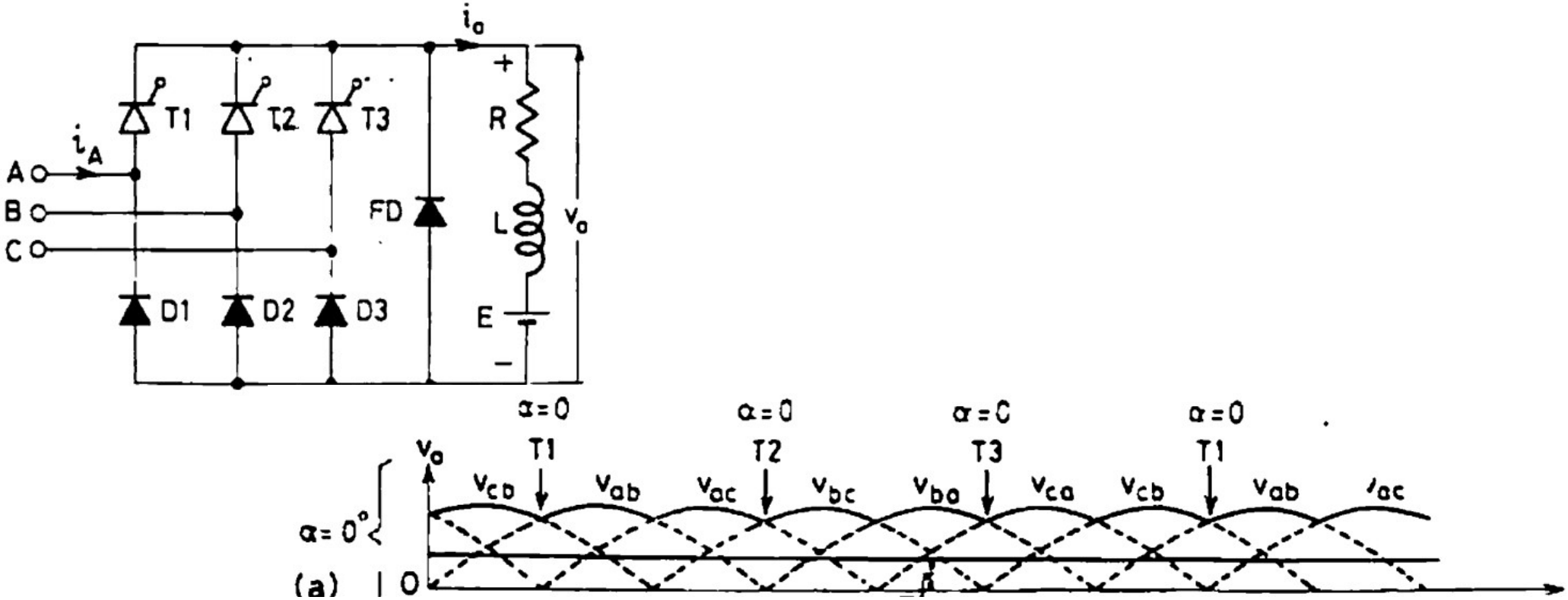


Three phase semi converter



The output voltage v_o across the load terminals is controlled by varying the firing angles of SCRs T_1, T_2 and T_3 .

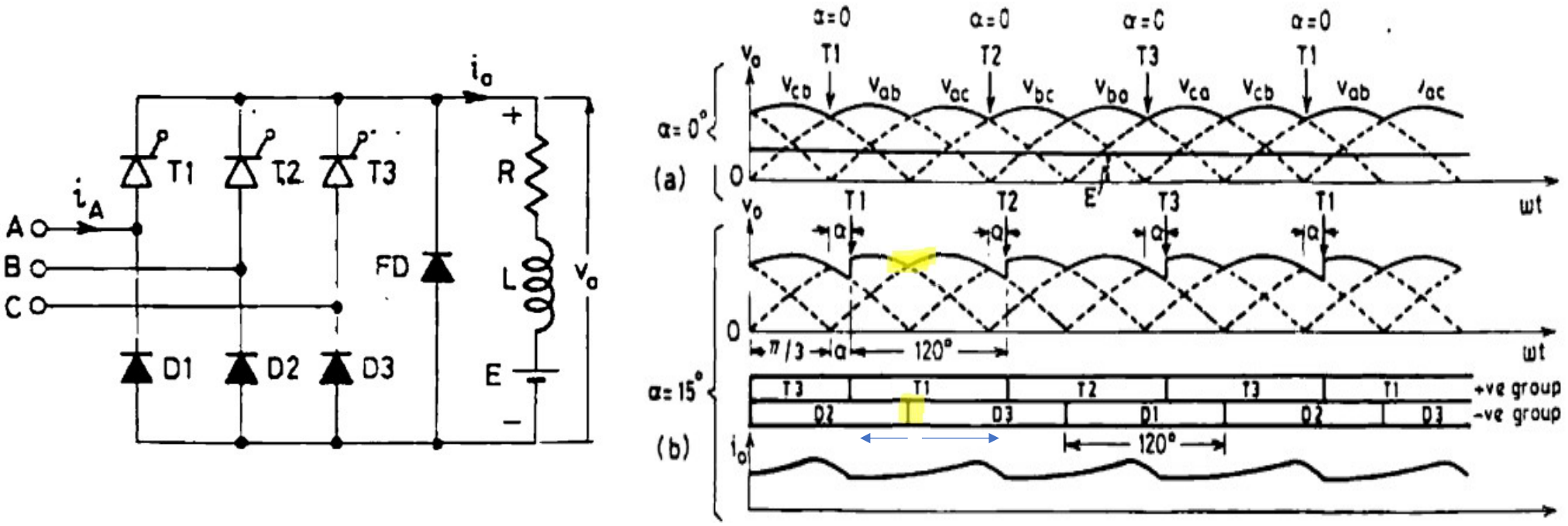
The diodes D_1, D_2 and D_3 provide merely a return path for the current through the most negative line terminal.



SIX PULSE CONVERTER (α <60o)

For a firing angle delay of α=0∘, thyristors $T1, T2, T3,$ would behave as diodes and the output voltage of the semi converter would be symmetrical, six pulses per cycle as shown.

The output voltage consisting of pulses Vab, Vac, Vbc, Vba, Vca, Vcb as shown.

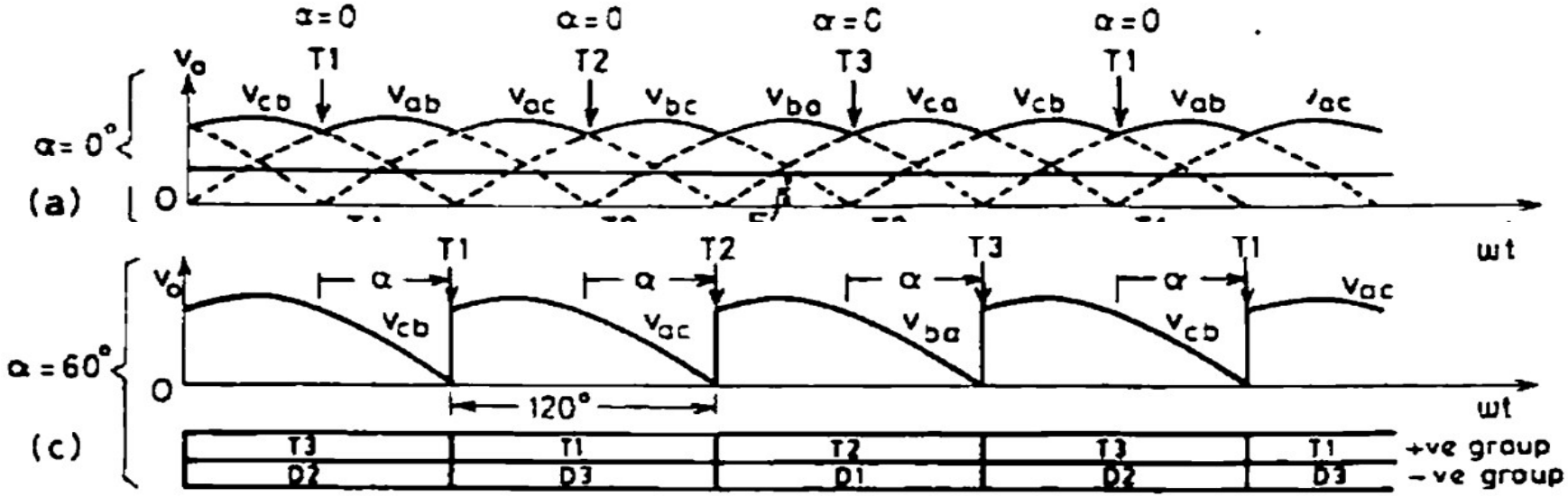


When the firing angle is delayed to α=15∘ , the triggering of SCRs $T1$, $T2$, $T3$, is delayed but return diodes $D1$,$D2$,$D3$ remain unaffected so that only alternate pulses are altered.

The load current is continuous and has little ripple.

The freewheeling diode (FD) does not come into play for α=15∘.

Each SCR and diode conduct for 120∘.



THREE PULSE CONVERTER(α>=60∘)

For α=60∘, the thyristors are fired so that current returns through one diode during each 120∘conduction period.

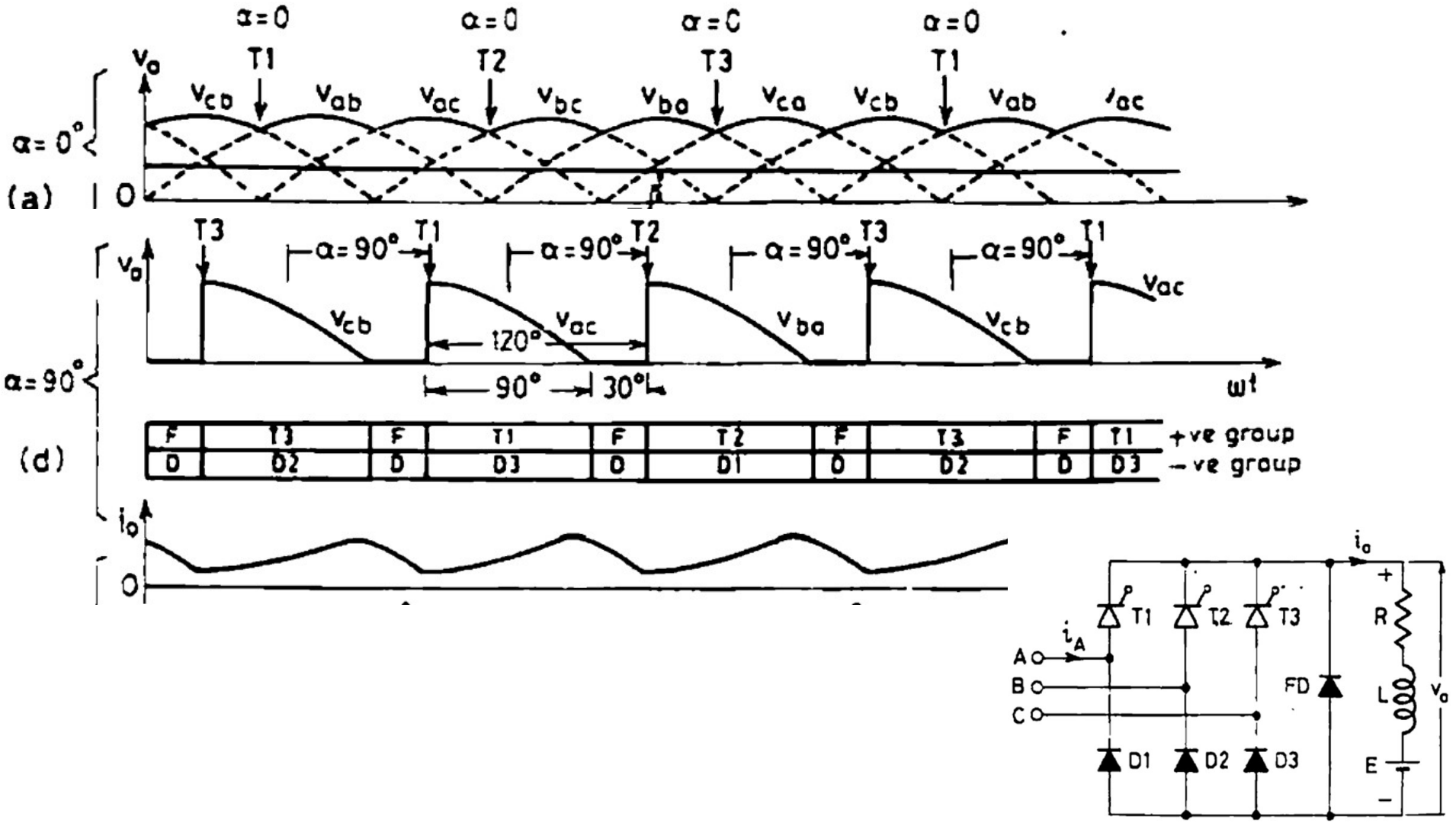
For voltage Vab, $T1$ and $D3$ conduct simultaneously for 120∘.

Similarly, other elements conduct.

The freewheeling diode (FD) does not come into play even for α=60∘.

It will be seen that for α≥60∘, voltage pulses Vab, Vbc, Vca are not present at output.

The load current, assumed continuous for α=60∘.


α = 0
α = 0
α = 0
α = 0
T1
T2
T3
T1
V_o
V_{cb}
V_{ab}
V_{ac}
V_{bc}
V_{ba}
V_{ca}
V_{cb}
V_{ab}
V_{ac}
α = 0°
(a)
O
T3
α = 90°
T1
α = 90°
T2
α = 90°
T3
α = 90°
T1
V_{ac}
V_{cb}
V_{ac}
V_{ba}
V_{cb}
120°
α = 90°
90°
30°
ωt
F
T3
F
T1
F
T2
F
T3
F
T1
+ve group
D
D2
D
D3
D
D1
D
D2
D
D3
−ve group
(d)
i_o
O
i_o
T1
T2
T3
i_A
A
B
C
FD
R
L
V_o
D1
D2
D3
E
+
−

- For a firing angle delay of 90°, the output voltage V0 is discontinuous.

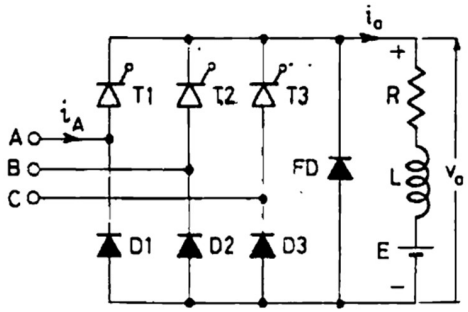
- As V0v0 , made up of Vcb,Vac,Vba, ,…, tends to become negative at ωt=120∘,240∘,360∘ωt=120∘,240∘,360∘, the freewheeling diode (FD) gets forward biased.

- Therefore, for each periodic cycle of 120°, the output voltage is equal to the line voltage for only 90°, and for the remaining 30°, when FD conducts, V0=0.

- For α=90∘, the conduction angle of SCRs and diodes is seen to be less than 120° for every output pulse.

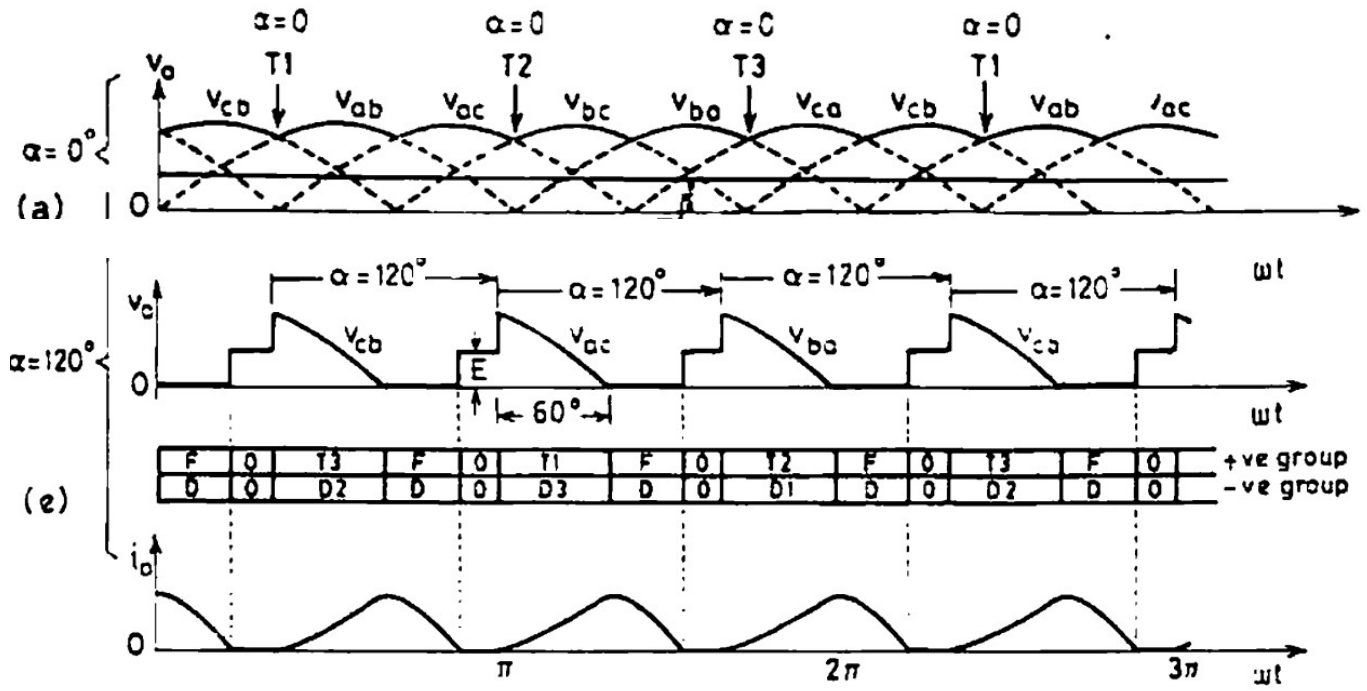
- In other words, the conduction angle for both positive and negative group thyristors is 90°, and for the remaining 30°, current completes its path through FD.

- Voltage pulses vab, vbc, vca are absent from output voltage v0 for this firing angle as well.

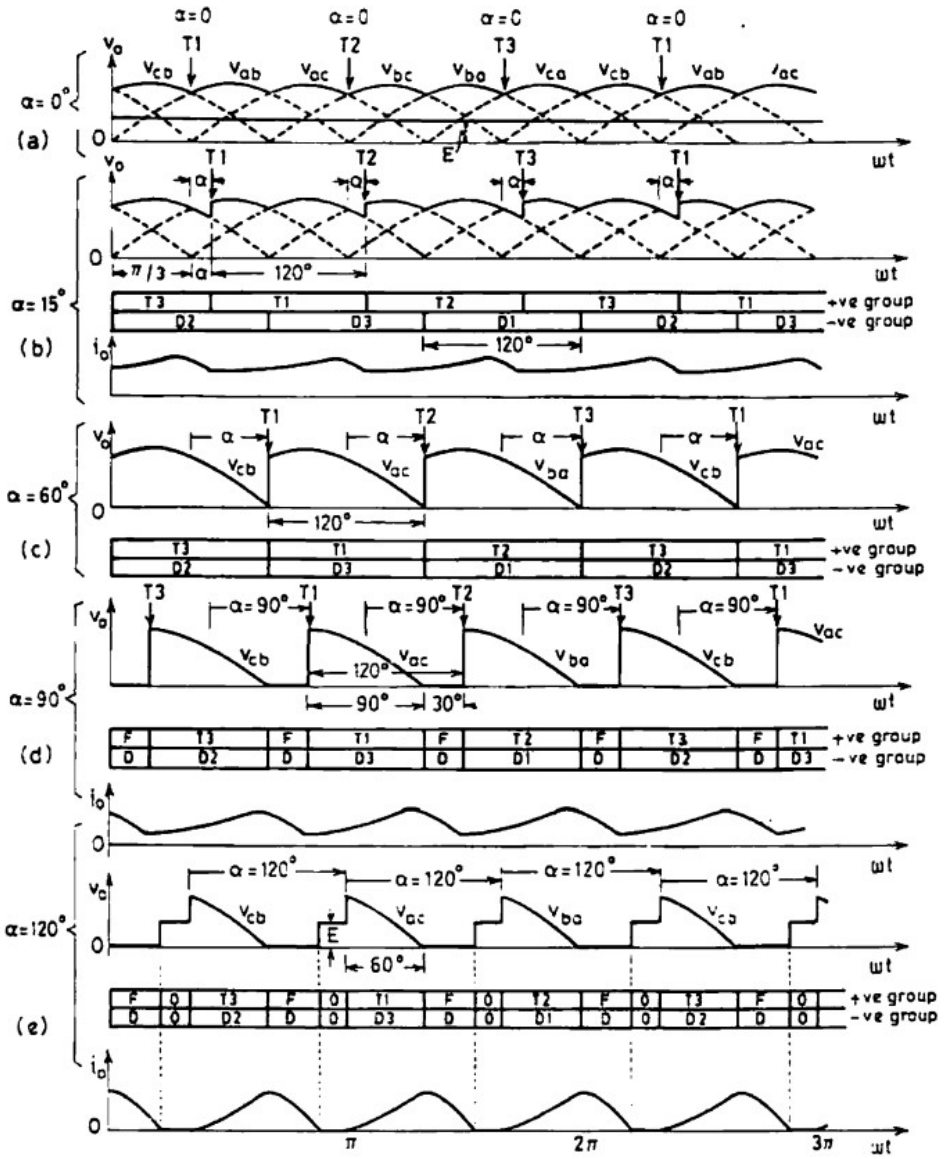


- Without FD, after the load voltage v0 reaches zero, a diode from the negative group would begin to conduct, reducing v0 to zero till the next SCR in sequence is triggered.

- For example, at ωt=120∘, V0=Vcb , and without FD, D3 from the negative group would start conducting through T3 from ωt=120∘to 150∘ when SCR T1 is gated.

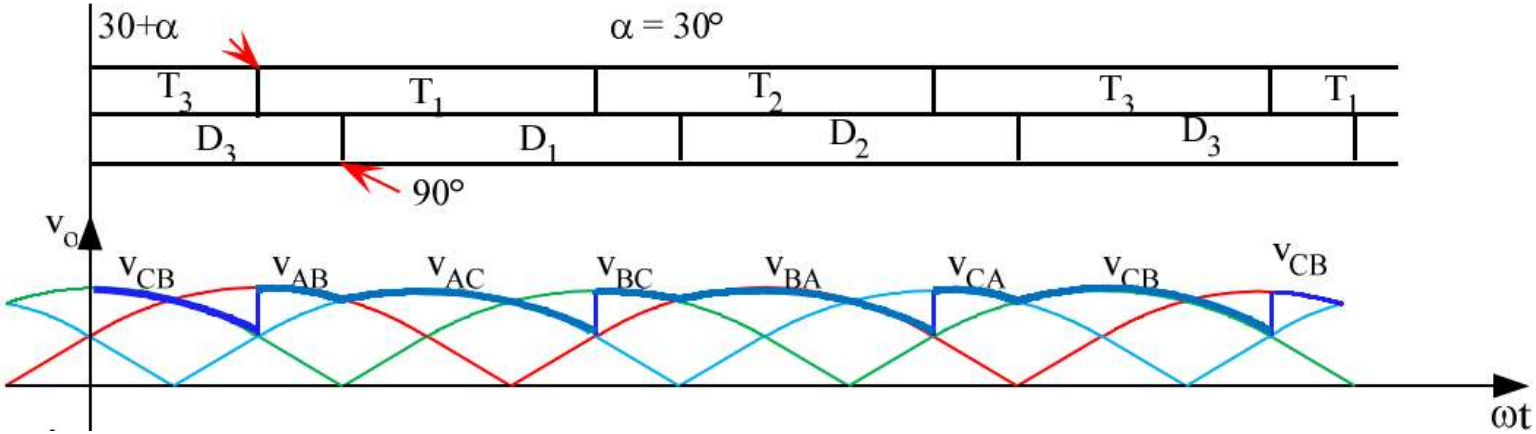

α=0
α=0
α=0
α=0
T1
T2
T3
T1
V_o
V_{cb}
V_{ab}
V_{ac}
V_{bc}
V_{ba}
V_{ca}
V_{cb}
V_{ab}
V_{ac}
α= 0°
(a)
O
α=120°
α=120°
α=120°
α=120°
ωt
V_o
V_{cb}
V_{ac}
V_{ba}
V_{ca}
α=120°
O
E
60°
ωt
F O T3 F O T1 F O T2 F O T3 F O
D O D2 D O D3 D O D1 D O D2 D O
+ve group
-ve group
(e)
i_o
O
π
2π
3π
ωt

- For a firing angle delay of 120°, the voltage and load current are assumed discontinuous.
- For each periodic cycle of 120°, v_0 is seen to have three components.
- When an SCR is gated, the thyristor and diode conduct for 60° only.
- As V_0 reaches zero and tends to become negative, the freewheeling diode (FD) gets forward biased and therefore starts conducting for some angle and holds the load voltage to zero.
- When all the energy stored in inductance is discharged, FD stops conducting and as a result, load voltage rises to the load counter emf E.
- When $v_0 = E$, none of the elements of the semi converter bridge is conducting.

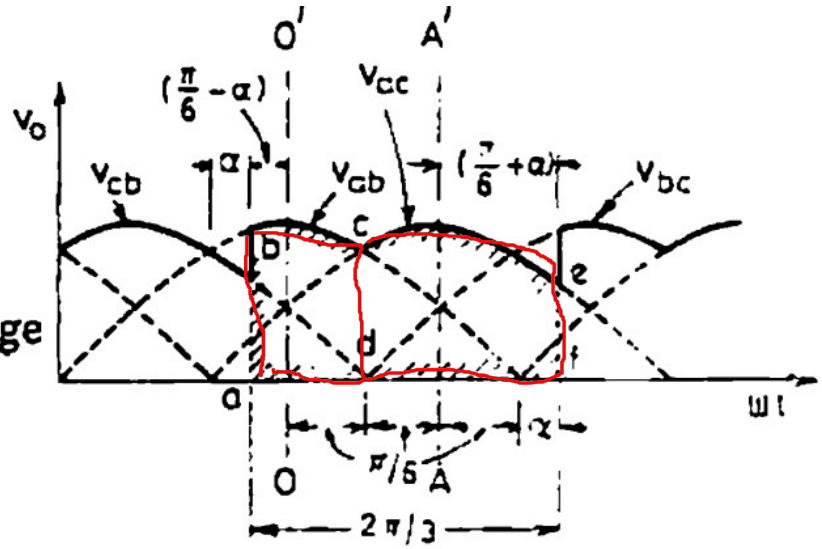

α = 0
T1
T2
T3
Vo
Vcb
Vab
Vac
Vbc
Vba
Vca
E
ωt
α = 0°
(a)
π/3
120°
α = 15°
(b)
+ve group
-ve group
io
α = 60°
(c)
α = 90°
(d)
F
D
D2
D3
D1
α = 120°
60°
(e)
π
2π
3π

AVERAGE OUTPUT VOLTAGE

For $\alpha \leq 60°$, during one pulse, T_1D_3 and T_1D_1 conducts, both $v_o = v_{AB}$ and $v_o = v_{AC}$ occur.
For $\alpha > 60°$, for one pulse, T_1D_1 and T_1D_2 conducts, $v_o = v_{AC}$ only occurs.



Average output voltage



For $\alpha < 60°$. For firing angle less than 60°, the output voltage

$$V_0 = \frac{3}{2\pi}\,[\text{Area } abcda + \text{Area } dcefd]$$

$$= \frac{3}{2\pi}\left[\int_{-\left[\frac{\pi}{6}-\alpha\right]}^{\pi/6} V_{ml}\cos\omega t\cdot d(\omega t) + \int_{-\pi/6}^{\left(\frac{\pi}{6}+\alpha\right)} V_{ml}\cos\omega t\cdot d(\omega t)\right]$$

$$V_0 = \frac{3\,V_{m.}}{2\pi}\,(1+\cos\alpha)$$

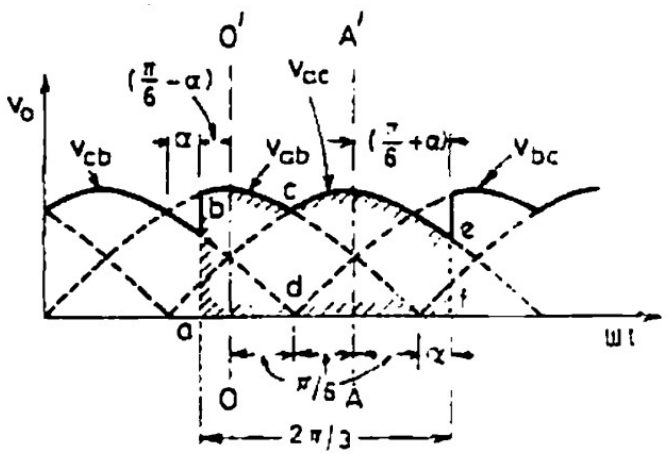
With OO' as the origin, angle $\left(\frac{\pi}{6}-\alpha\right)$ is measured to the left of OO', therefore minus sign is put before $\left(\frac{\pi}{6}-\alpha\right)$. Similarly, minus sign is put before $\frac{\pi}{6}$.

Alternately....

Voltage $v_{ab} = 0$ at $\omega t = 0$ and $v_{ac} = 0$ at $\omega t = \frac{\pi}{3}$

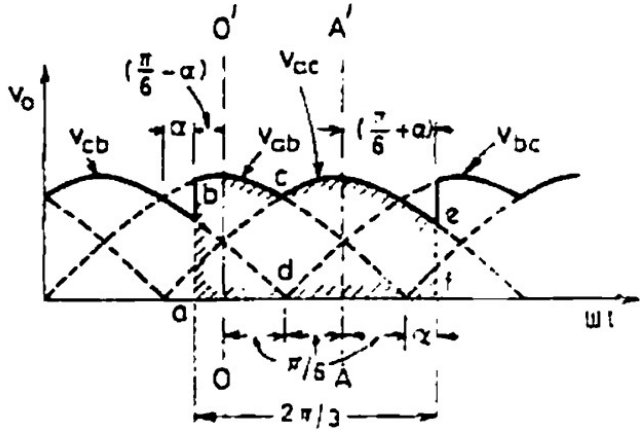
$$V_0 = \frac{3}{2\pi}\left[\int_{\frac{\pi}{3}+\alpha}^{2\pi/3} V_{ml} \sin \omega t \, d(\omega t) + \int_{\pi/3}^{\frac{2\pi}{3}+\alpha} V_m \sin \omega t \, d(\omega t)\right]$$

$$= \frac{3\,V_{ml}}{2\pi}(1 + \cos \alpha)$$



Rms value of output voltage V_{or} for $\alpha < 60°$, is given by

RMS value of output



Rms value of output voltage V_{or}, for $\alpha < 60°$, is given by

$$V_{or} = \left[\frac{3}{2\pi} \left\{ \int_{-\left(\frac{\pi}{6}-\alpha\right)}^{\pi/6} V_{ml}^2 \cos^2 \omega t \, d(\omega t) + \int_{\pi/6}^{\frac{\pi}{6}+\alpha} V_{ml}^2 \cos^2 \omega t \, d(\omega t) \right\} \right]^{1/2}$$

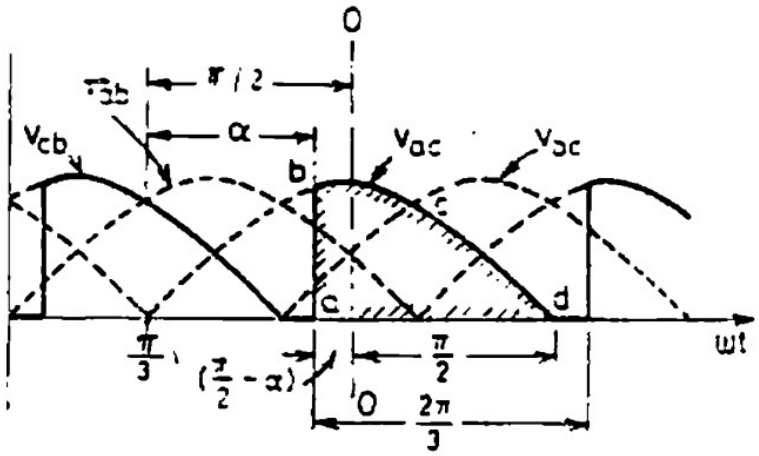
$$V_{or}^2 = \frac{3 V_{ml}^2}{4\pi} \left[\left| \omega t + \frac{\sin 2\omega t}{2} \right|_{-\left(\frac{\pi}{6}-\alpha\right)}^{\pi/6} + \left| \omega t + \frac{\sin 2\omega t}{2} \right|_{-\pi/6}^{\frac{\pi}{6}+\alpha} \right]$$

$$V_{or} = \frac{V_{ml}}{2} \sqrt{\frac{3}{\pi}} \left[\frac{2\pi}{3} + \frac{\sqrt{3}}{2} (1 + \cos 2\alpha) \right]^{1/2} \qquad \ldots(6.56)$$

For $\alpha \geq 60°$. For $\alpha \geq 60°$, the output voltage waveform is drawn in Fig. 6.32 (*b*) for a firing angle $60° < \alpha < 90°$ for convenience. With OO' as the origin in this figure, the average output voltage V_0 is given by

$$V_0 = \frac{3}{2\pi} [\text{Area } abcda].$$

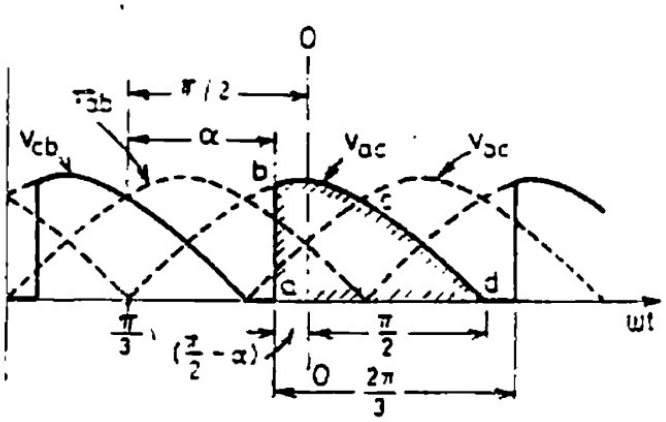
$$= \frac{3}{2\pi} \left[\int_{-\left[\frac{\pi}{2} - \alpha\right]}^{\pi/2} V_{ml} \cos \omega t \cdot d(\omega t) = \frac{3\, V_{ml}}{2\pi} (1 + \cos \alpha) \right.$$



3-phase semiconverter can operate only in one quadrant since average load voltage is always positive for all values of α.

Average out put voltage is same for the six pulse and three pulse operation

RMS value of 3 pulse converter

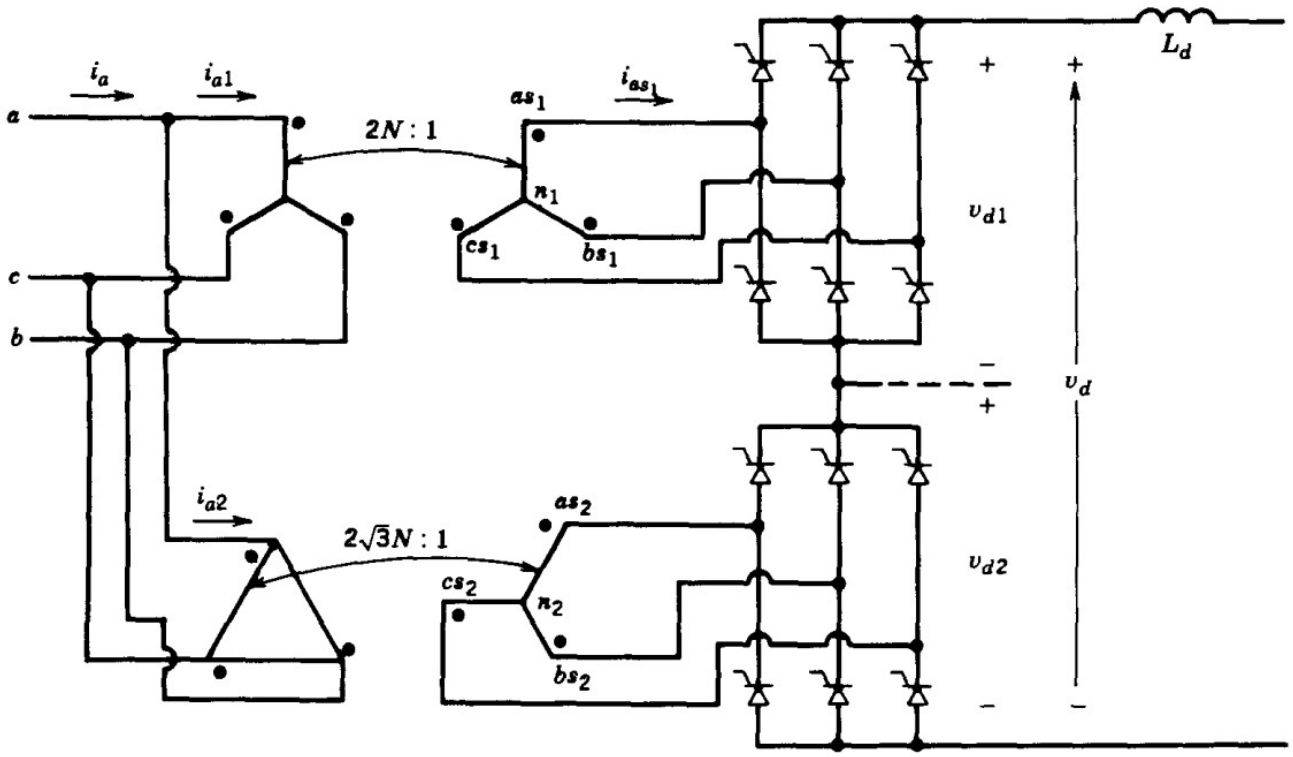


Rms value of output voltage V_{or}, for $\alpha > 60°$ is given by

$$V_{or} = \left[\frac{3}{2\pi} \int_{-\left(\frac{\pi}{2}-\alpha\right)}^{\pi/2} V_{ml}^2 \cos^2 \omega t \, . \, d(\omega t) \right]^{1/2}$$

$$= \frac{V_{ml}}{2} \cdot \sqrt{\frac{3}{\pi}} \left[(\pi - \alpha) + \frac{1}{2} \sin 2\alpha \right]^{1/2}$$

12 pulse converter – Series connection



The series connection of two 6-pube converters on the dc side is important to meet the high voltage requirement of an HVDC system.

- D-Y and Y -Y transformers in order to obtain a 12-pulse converter and a reduced output ripple.

- The filters are required to reduce the current harmonic generated by the converter.

- 12-pulse bridge converters are used for high-voltage dc transmission

Parallel connection of two 6-pulse converters for high-current applications

